



1. Description

1.1. Project

Project Name	AB LSTM
Board Name	NUCLEO-N657X0-Q
Generated with:	STM32CubeMX 6.15.0
Date	08/12/2025

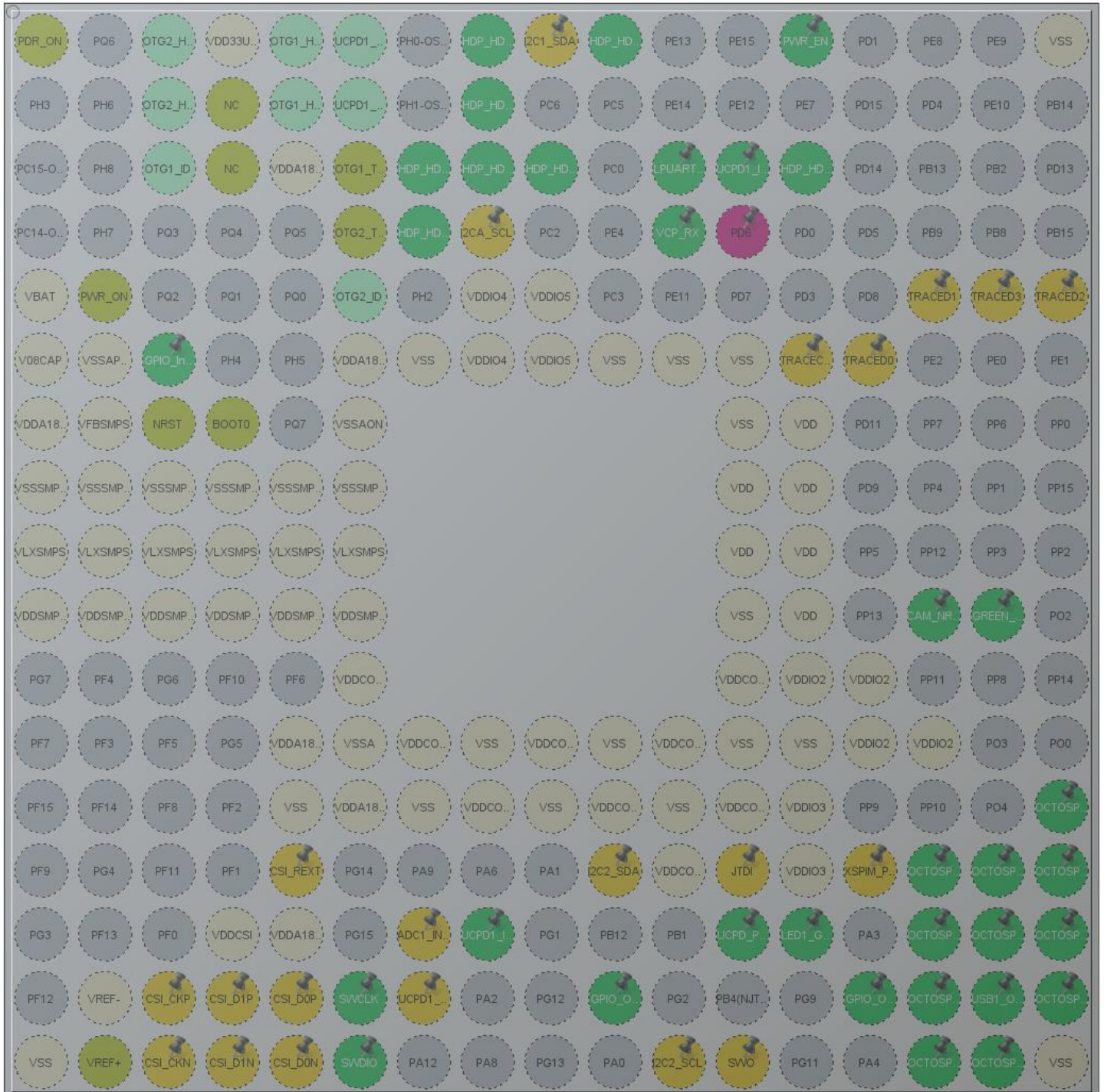
1.2. MCU

MCU Series	STM32N6
MCU Line	STM32N6x7
MCU name	STM32N657X0HxQ
MCU Package	VFBGA264
MCU Pin number	264

1.3. Core(s) information

Core(s)	ARM Cortex-M55
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2. Pinout Configuration



VFBGA264 (Top view)

3. Pins Configuration

Pin Number VFBGA264	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A1	PDR_ON	MonoIO		
A4	VDD33USB	Power		
A8	PC10	I/O	HDP_HDP2	
A9	PC1 *	I/O	I2C1_SDA	I2C1_SDA
A10	PC4	I/O	HDP_HDP6	
A13	PD10 **	I/O	GPIO_Output	PWR_EN
A17	VSS	Power		
B4	NC	NC		
B8	PC11	I/O	HDP_HDP3	
C4	NC	NC		
C5	VDDA18USB	Power		
C6	OTG1_TXRTUNE	MonoIO		
C7	PC9	I/O	HDP_HDP1	
C8	PC12	I/O	HDP_HDP4	
C9	PC7	I/O	HDP_HDP7	
C11	PE5	I/O	LPUART1_TX	
C12	PD2 **	I/O	GPIO_Input	UCPD1_INT
C13	PD12	I/O	HDP_HDP5	
D6	OTG2_TXRTUNE	MonoIO		
D7	PC8	I/O	HDP_HDP0	
D8	PH9 *	I/O	I2C1_SCL	I2CA_SCL
D11	PE6	I/O	LPUART1_RX	VCP_RX
D12	PD6	I/O		
E1	VBAT	Power		
E2	PWR_ON	MonoIO		
E8	VDDIO4	Power		
E9	VDDIO5	Power		
E15	PB0 *	I/O	DEBUG_TRACED1	TRACED1
E16	PB7 *	I/O	DEBUG_TRACED3	TRACED3
E17	PB6 *	I/O	DEBUG_TRACED2	TRACED2
F1	V08CAP	Power		
F2	VSSAPMU	Power		
F3	PC13 **	I/O	GPIO_Input	
F6	VDDA18AON	Power		
F7	VSS	Power		
F8	VDDIO4	Power		

Pin Number VFBGA264	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
F9	VDDIO5	Power		
F10	VSS	Power		
F11	VSS	Power		
F12	VSS	Power		
F13	PB3 *	I/O	DEBUG_TRACECLK	TRACECLK
F14	PE3 *	I/O	DEBUG_TRACED0	TRACED0
G1	VDDA18PMU	Power		
G2	VFBSMPS	Power		
G3	NRST	Reset		
G4	BOOT0	Boot		
G6	VSSAON	Power		
G12	VSS	Power		
G13	VDD	Power		
H1	VSSSMPS	Power		
H2	VSSSMPS	Power		
H3	VSSSMPS	Power		
H4	VSSSMPS	Power		
H5	VSSSMPS	Power		
H6	VSSSMPS	Power		
H12	VDD	Power		
H13	VDD	Power		
J1	VLXSMPS	Power		
J2	VLXSMPS	Power		
J3	VLXSMPS	Power		
J4	VLXSMPS	Power		
J5	VLXSMPS	Power		
J6	VLXSMPS	Power		
J12	VDD	Power		
J13	VDD	Power		
K1	VDDSMPS	Power		
K2	VDDSMPS	Power		
K3	VDDSMPS	Power		
K4	VDDSMPS	Power		
K5	VDDSMPS	Power		
K6	VDDSMPS	Power		
K12	VSS	Power		
K13	VDD	Power		
K15	PO5 **	I/O	GPIO_Output	CAM_NRST
K16	PO1 **	I/O	GPIO_Output	GREEN_LED

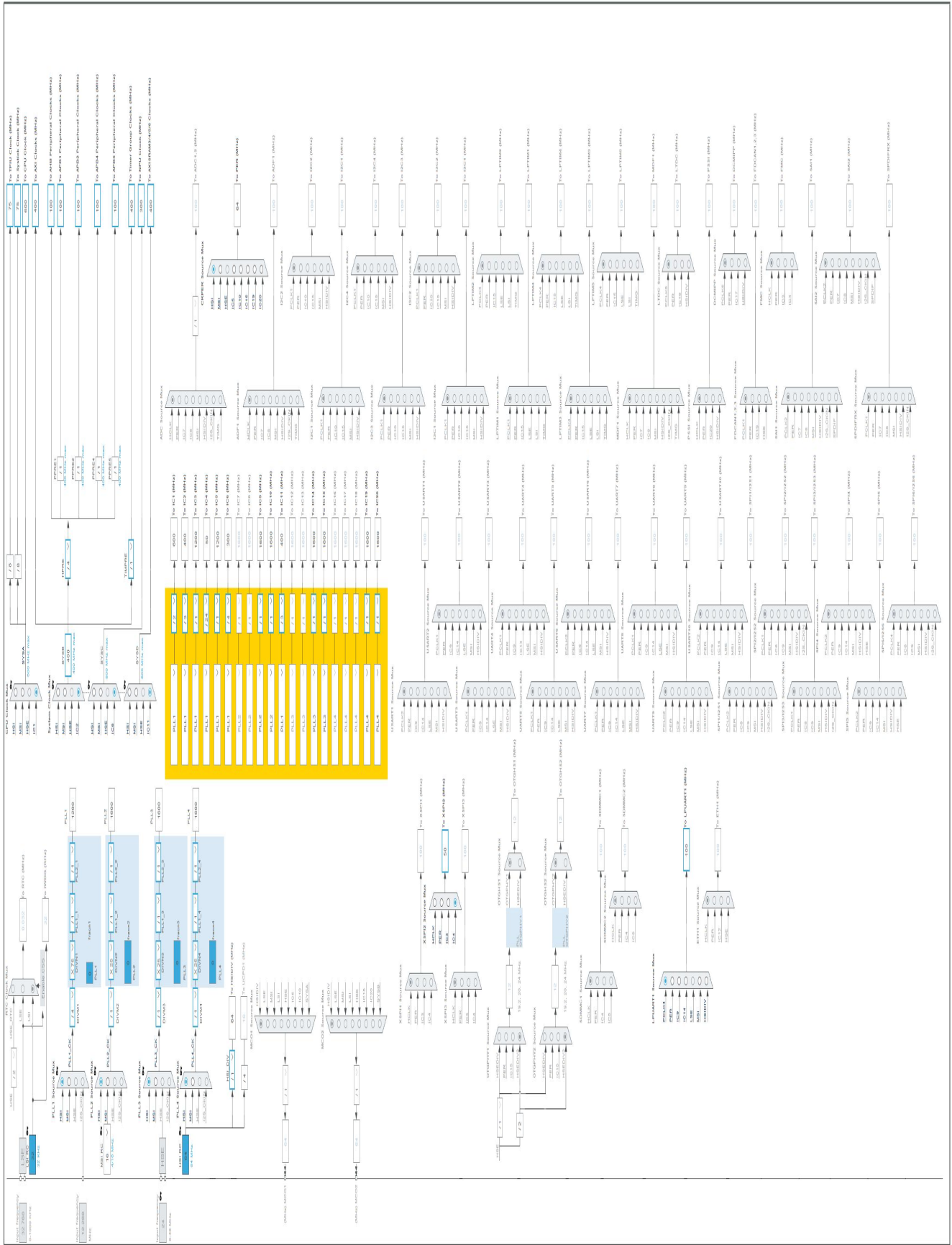
Pin Number VFBGA264	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
L6	VDDCORE	Power		
L12	VDDCORE	Power		
L13	VDDIO2	Power		
L14	VDDIO2	Power		
M5	VDDA18PLL	Power		
M6	VSSA	Power		
M7	VDDCORE	Power		
M8	VSS	Power		
M9	VDDCORE	Power		
M10	VSS	Power		
M11	VDDCORE	Power		
M12	VSS	Power		
M13	VSS	Power		
M14	VDDIO2	Power		
M15	VDDIO2	Power		
N5	VSS	Power		
N6	VDDA18ADC	Power		
N7	VSS	Power		
N8	VDDCORE	Power		
N9	VSS	Power		
N10	VDDCORE	Power		
N11	VSS	Power		
N12	VDDCORE	Power		
N13	VDDIO3	Power		
N17	PN4	I/O	XSPIM_P2_IO2	OCTOSPI_IO2
P5	CSI_REXT *	MonoIO	CSI_REXT	
P10	PB11 *	I/O	I2C2_SDA	I2C2_SDA
P11	VDDCORE	Power		
P12	PA15(JTDI) *	I/O	DEBUG_JTDI	JTDI
P13	VDDIO3	Power		
P14	PN12 *	I/O	XSPIM_P2_NCS2	
P15	PN6	I/O	XSPIM_P2_CLK	OCTOSPI_CLK
P16	PN8	I/O	XSPIM_P2_IO4	OCTOSPI_IO4
P17	PN0	I/O	XSPIM_P2_DQS0	OCTOSPI_DQS
R4	VDDCSI	Power		
R5	VDDA18CSI	Power		
R7	PA10 *	I/O	ADC1_INP11	
R8	PA5 **	I/O	GPIO_Output	UCPD1_ISENSE
R12	PA7 **	I/O	GPIO_Output	UCPD_PWR_EN

Pin Number VFBGA264	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
R13	PG0 **	I/O	GPIO_Output	LED1_GREEN
R15	PN3	I/O	XSPIM_P2_IO1	OCTOSPI_IO1
R16	PN5	I/O	XSPIM_P2_IO3	OCTOSPI_IO3
R17	PN1	I/O	XSPIM_P2_NCS1	OCTOSPI_NCS
T2	VREF-	Power		
T3	CSI_CKP *	MonoIO	CSI_CKP	
T4	CSI_D1P *	MonoIO	CSI_D1P	
T5	CSI_D0P *	MonoIO	CSI_D0P	
T6	PA14(JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	SWCLK
T7	PA11 *	I/O	ADC1_INP12, ADC1_INN11	UCPD1_VSENSE
T10	PG10 **	I/O	GPIO_Output	
T14	PG8 **	I/O	GPIO_Output	
T15	PN9	I/O	XSPIM_P2_IO5	OCTOSPI_IO5
T16	PN7 **	I/O	GPIO_Output	USB1_OCP
T17	PN2	I/O	XSPIM_P2_IO0	OCTOSPI_IO0
U1	VSS	Power		
U2	VREF+	MonoIO		
U3	CSI_CKN *	MonoIO	CSI_CKN	
U4	CSI_D1N *	MonoIO	CSI_D1N	
U5	CSI_D0N *	MonoIO	CSI_D0N	
U6	PA13(JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	SWDIO
U11	PB10 *	I/O	I2C2_SCL	I2C2_SCL
U12	PB5(JTDO/TRACESWO) *	I/O	DEBUG_JTDO-SWO	SWO
U15	PN10	I/O	XSPIM_P2_IO6	OCTOSPI_IO6
U16	PN11	I/O	XSPIM_P2_IO7	OCTOSPI_IO7
U17	VSS	Power		

** The pin is affected with an I/O function

* The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32N6
Line	STM32N6x7
MCU	STM32N657X0HxQ
Datasheet	DS00002 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	1.8

1.3. Battery Selection

Battery	Li-SOCL2(DD36000)
Capacity	36000.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	450.0 mA
Max Pulse Current	1000.0 mA
Cells in series	1
Cells in parallel	1

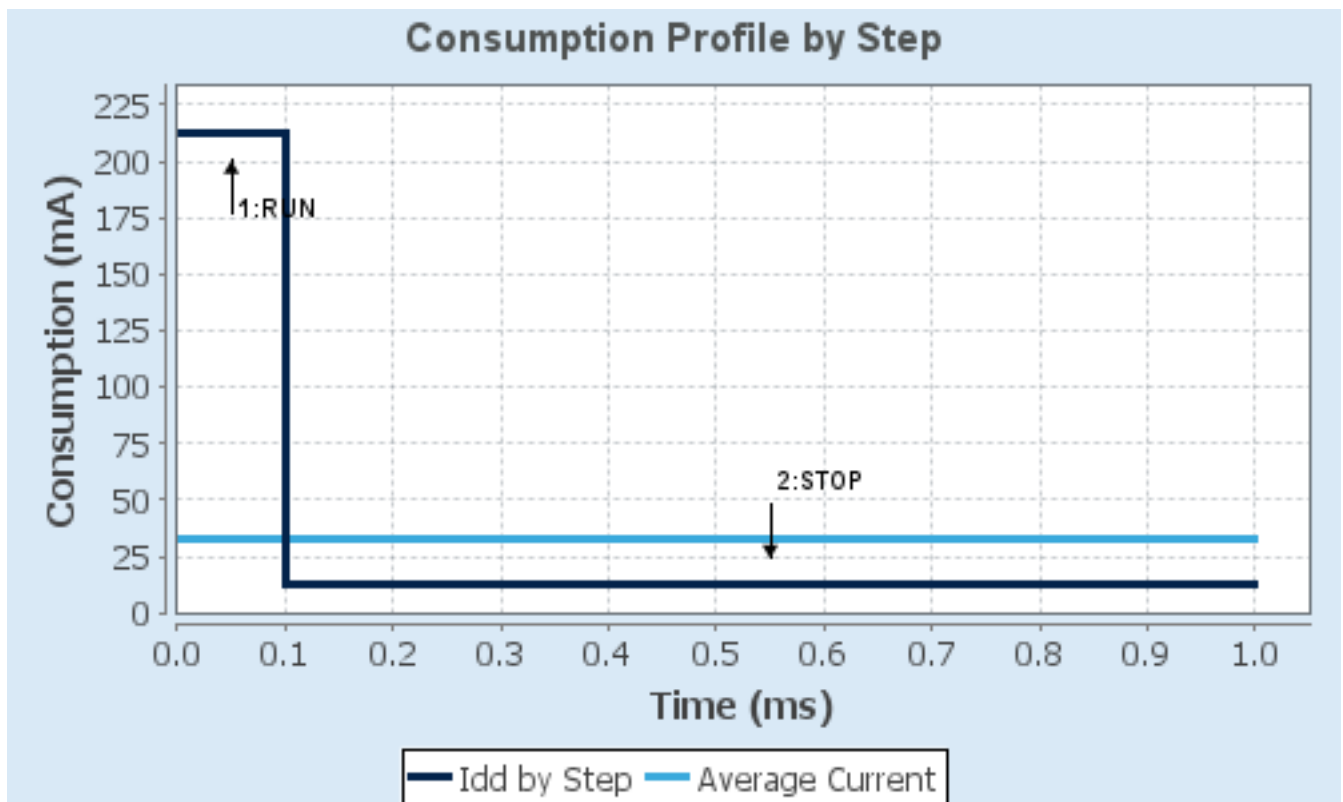
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	1.8	1.8
Voltage Source	Battery	Battery
Range	VOS_HIGH/extern SMPS	SVOS_LOW/intern SMPS
Fetch Type	AXI-SRAM2/CACHE-ON	FLASH
CPU Frequency	800 Hz	0 Hz
Clock Configuration	HSI PLL:400/800 ALL_IPs_ON	ALL CLOCKS OFF
Clock Source Frequency	400 Hz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	212.16 mA	12.3 mA
Duration	0.1 ms	0.9 ms
DMIPS	1.0	0.0
Ta Max	90.25	122.99
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	32.29 mA
Battery Life	1 month, 15 days, 23 hours	Average DMIPS	0.10239999 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	AB_LSTM
Project Folder	C:\ST\AB_LSTM\AB_LSTM
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_N6 V1.2.0
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x800

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls FSBL

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_ICACHE_Init	ICACHE
4	MX_XSPIM_Init	XSPIM
5	MX_HDP_Init	HDP
6	MX_LPUART1_UART_Init	LPUART1
7	MX_XSPI2_Init	XSPI2
8	MX_EXTMEM_MANAGER_Init	EXTMEM_MANAGER

2.4. Advanced Settings - Generated Function Calls Appli

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	MX_LPUART1_UART_Init	LPUART1
3	MX_XSPI2_Init	XSPI2
4	MX_XSPIM_Init	XSPIM
5	SystemIsolation_Config	RIF

2.5. Advanced Settings - Generated Function Calls ExtMemLoader

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	MX_XSPIM_Init	XSPIM
3	MX_XSPI2_Init	XSPI2
4	MX_EXTMEM_MANAGER_Init	EXTMEM_MANAGER

3. Peripherals and Middlewares Configuration

3.1. CORTEX_M55_FSBL

3.1.1. Parameter Settings:

Core(s) Settings:

Context(s): First Stage Boot Loader

Cortex Interface Settings:

CPU ICache Enabled *

CPU DCache Enabled *

Cortex Memory Protection Unit Control Settings:

MPU Control Mode Background Region Privileged accesses only + MPU Enabled during hard fault, NMI and FAULTMASK handlers *

Cortex Memory Protection Unit Region 0 Settings:

Region Enabled *

Region Base Address 0x70000000 *

Region Limit Address 0x73FFFFFF *

Attributes Number ATTRIBUTE 0

Access Permission ALL READS/Writes Permissions *

Instruction Access Enable

Privileged Instruction Access Enable

Shareability Permission Inner Shareable *

Cortex Memory Protection Unit Region 1 Settings:

Region Disabled

Cortex Memory Protection Unit Region 2 Settings:

Region Disabled

Cortex Memory Protection Unit Region 3 Settings:

Region Disabled

Cortex Memory Protection Unit Region 4 Settings:

Region Disabled

Cortex Memory Protection Unit Region 5 Settings:

Region Disabled

Cortex Memory Protection Unit Region 6 Settings:

Region Disabled

Cortex Memory Protection Unit Region 7 Settings:

Region Disabled

Cortex Memory Protection Unit Region 8 Settings:

Region Disabled

Cortex Memory Protection Unit Region 9 Settings:

Region Disabled

Cortex Memory Protection Unit Region 10 Settings:

Region Disabled

Cortex Memory Protection Unit Region 11 Settings:

Region Disabled

Cortex Memory Protection Unit Region 12 Settings:

Region Disabled

Cortex Memory Protection Unit Region 13 Settings:

Region Disabled

Cortex Memory Protection Unit Region 14 Settings:

Region Disabled

Cortex Memory Protection Unit Region 15 Settings:

Region Disabled

3.1.2. Attributes Settings:

Core(s) Settings:

Context(s): First Stage Boot Loader

Cortex Memory Protection Unit Attributes 0 Settings:

Attributes Enabled *
Memory type Normal cacheable *
Cache policy WRITE BACK *
Transient Permission NOT TRANSIENT *
Allocate Permission R ALLOCATE *

Cortex Memory Protection Unit Attributes 1 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 2 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 3 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 4 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 5 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 6 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 7 Settings:

Attributes Disabled

3.2. CORTEX_M55_S

3.2.1. Parameter Settings:

Core(s) Settings:

Context(s): Application

Cortex Interface Settings:

CPU ICache Enabled *

CPU DCache Enabled *

Cortex Memory Protection Unit Control Settings:

MPU Control Mode Background Region Privileged accesses only + MPU Enabled during hard fault, NMI and FAULTMASK handlers *

Cortex Memory Protection Unit Region 0 Settings:

Region Enabled *

Region Base Address 0x0

Region Limit Address 0x0

Attributes Number ATTRIBUTE 1

Access Permission ALL READS/Writes Permissions *

Instruction Access Enable

Privileged Instruction Access Enable

Shareability Permission Disable

Cortex Memory Protection Unit Region 1 Settings:

Region Disabled

Cortex Memory Protection Unit Region 2 Settings:

Region Disabled

Cortex Memory Protection Unit Region 3 Settings:

Region Disabled

Cortex Memory Protection Unit Region 4 Settings:

Region Disabled

Cortex Memory Protection Unit Region 5 Settings:

Region Disabled

Cortex Memory Protection Unit Region 6 Settings:

Region Disabled

Cortex Memory Protection Unit Region 7 Settings:

Region Disabled

Cortex Memory Protection Unit Region 8 Settings:

Region Disabled

Cortex Memory Protection Unit Region 9 Settings:

Region Disabled

Cortex Memory Protection Unit Region 10 Settings:

Region Disabled

Cortex Memory Protection Unit Region 11 Settings:

Region Disabled

Cortex Memory Protection Unit Region 12 Settings:

Region Disabled

Cortex Memory Protection Unit Region 13 Settings:

Region Disabled

Cortex Memory Protection Unit Region 14 Settings:

Region Disabled

Cortex Memory Protection Unit Region 15 Settings:

Region Disabled

3.2.2. Attributes Settings:

Core(s) Settings:

Context(s): Application

Cortex Memory Protection Unit Attributes 0 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 1 Settings:

Attributes **Enabled ***

Memory type Normal non-cacheable

Cortex Memory Protection Unit Attributes 2 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 3 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 4 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 5 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 6 Settings:

Attributes Disabled

Cortex Memory Protection Unit Attributes 7 Settings:

Attributes Disabled

3.3. DEBUG

Debug: Serial Wire

3.3.1. Core(s) Settings:

Context(s): First Stage Boot Loader
Application

3.4. HDP

mode: Hardware Debug Port

3.4.1. Core(s) Settings:

Context(s): First Stage Boot Loader

3.5. ICACHE

Mode: 1-way (direct mapped cache)

Secure Attribute: not secured

3.5.1. Core(s) Settings:

Context(s): First Stage Boot Loader

3.6. LPUART1

Mode: Asynchronous

3.6.1. Parameter Settings:

Core(s) Settings:

Context(s): First Stage Boot Loader
Application

Basic Parameters:

Baud Rate 921600 *
Word Length 8 Bits (including Parity)

Parity	None
Stop Bits	1
Advanced Parameters:	
Data Direction	Receive and Transmit
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration
Autonomous Mode	Disable

Advanced Features:

TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.7. PWR

mode: I/O voltage range selection

mode: Dead Battery Signals disabled

mode: Power saving mode

mode: Security and Privilege attributes

3.7.1. Parameter Settings:

Core(s) Settings:

Context(s):	First Stage Boot Loader
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I/O voltage range selection:

VDDIO2	(3.3)
VDDIO3	(1.8) *
VDDIO4	(3.3)
VDDIO5	(3.3)

3.7.2. PWR Privilege :

Core(s) Settings:

Context(s):	First Stage Boot Loader
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Privilege PWR:

Privilege of PWR Secure Items	Disable
Privilege of PWR Non-Secure Items	Disable

3.7.3. PWR Security:

Core(s) Settings:

Context(s):	First Stage Boot Loader
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Secure PWR:

Wake-Up 1 secure protection	Disable
Wake-Up 2 secure protection	Disable
Wake-Up 3 secure protection	Disable
Wake-Up 4 secure protection	Disable
Low power modes secure protection	Disable
Backup domain secure protection	Disable
Retention secure protection	Disable
Supply configuration and monitoring secure protection	Disable
Voltage USB Secure Protection	Disable

3.8. RCC

3.8.1. Parameter Settings:

Core(s) Settings:

Context(s):	First Stage Boot Loader Application External Memory Loader
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Power Parameters:

SupplySource	PWR_EXTERNAL_SOURCE_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1

Spread spectrum mode:

PLL1 CSG mode	Disabled
PLL2 CSG mode	Disabled
PLL3 CSG mode	Disabled
PLL4 CSG mode	Disabled

RCC Parameters:

HSI Calibration Value	16
MSI Calibration Value	16
TIM Group1 Prescaler Selection	Disabled

TIM Group2 Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

System Parameters:

VDD voltage (V)	3.3
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3.9. RIF

mode: RISUP

3.9.1. Parameter Settings:

Core(s) Settings:

Context(s):	Application
0	
SPI1/I2S1	false
	false
	-
	false
1	
SPI2/I2S2	false
	false
	-
	false
2	
SPI3/I2S3	false
	false
	-
	false
3	
SPI4	false
	false
	-
	false
4	
SPI5	false
	false
	-
	false
5	
SPI6/I2S6	false
	false

	-
	false
6	
SAI1	false
	false
	-
	false
8	
SAI2	false
	false
	-
	false
9	
I2C1	false
	false
	-
	false
10	
I2C2	false
	false
	-
	false
11	
I2C3	false
	false
	-
	false
12	
I2C4	false
	false
	-
	false
13	
I3C1	false
	false
	-
	false
14	
I3C2	false
	false
	-
	false
15	

USART1	false
	false
	-
	false
16	
USART2	false
	false
	-
	false
17	
USART3	false
	false
	-
	false
18	
UART4	false
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UART5	false
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USART6	false
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UART7	false
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UART8	false
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23	
UART9	false
	false
	-

	false
24	
USART10	false
	false
	-
	false
25	
LPUART1	false
	false
	-
	false
26	
FDCAN1	false
	false
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	false
26	
FDCAN2	false
	false
	-
	false
26	
FDCAN3	false
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TIM1	false
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TIM2	false
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TIM4	false

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TIM11	false
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TIM12	false
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TIM13	false	
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TIM14	false	
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	-	
	false	
41		
TIM15	false	
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	-	
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42		
TIM16	false	
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43		
TIM17	false	
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44		
TIM18	false	
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45		
GEXTIM	false	
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	-	
	false	
46		
LPTIM1	false	
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	false	
47		
LPTIM2	false	
	false	

	-
	false
48	
LPTIM3	false
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LPTIM4	false
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50	
LPTIM5	false
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ADF1	false
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MDF1	false
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SDMMC1	false
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SDMMC2	false
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MDIOS	false
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OTG1_HS	false
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OTG2_HS	false
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UCPD1	false
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ETH1	false
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SPDIFRX	false
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SYSCFG	false
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VREFBUF	false
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CRC	false
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IWDG	false
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	false
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WWDG	false
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76	
RNG	false
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HASH	false
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80	
CRYP1	false
	false
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81	
MCE1	false
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82	
MCE2	false

	false
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83	
MCE3	false
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84	
MCE4	false
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	-
	false
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XSPI1	false
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XSPI2	false
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XSPI3	false
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89	
XSPIM	false
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90	
FMC	false
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CSI2HOST	false
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93		
DCMIPP	false	
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94		
DCMI	false	
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96		
JPEG	false	
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VENC	false	
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98		
ICACHE	false	
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	false	
99		
GPU	false	
	false	
	-	
	false	
100		
GFXMMU	false	
	false	
	-	
	false	
101		
DMA2D	false	
	false	
	-	
	false	
102		
LTDC_CMN	false	
	false	

	-
	false
103	
LTDC_L1	false
	false
	-
	false
104	
LTDC_L2	false
	false
	-
	false
106	
NPU	false
	false
	-
	false

3.10. XSPI2

Mode: Octo SPI

Port: Port2 Octo

Chip Select Override: NCS1 -- Port2 --

3.10.1. Parameter Settings:

Core(s) Settings:

Context(s):	First Stage Boot Loader
	Application
	External Memory Loader

Generic:

Fifo Threshold	4 *
Memory Mode	Disable
Memory Type	Macronix *
Memory Size	512 MBits *
Chip Select High Time Cycle	2 *
Free Running Clock	Disable
Clock Mode	Low
Wrap Size	Not Supported
Clock Prescaler	0

Sample Shifting	None
Delay Hold Quarter Cycle	Enable *
Chip Select Boundary	Disabled
Maximum Transfer	0
Refresh Rate	0
Memory Select	NCS1
Switching Duration Clock Number	1

3.11. XSPIM

Mode: Direct (XSPI1 to Port1; XSPI2 to Port2; XSPI3 not used)

3.11.1. Core(s) Settings:

Context(s):	First Stage Boot Loader Application External Memory Loader
-------------	--

3.12. EXTMEM_MANAGER

mode: Activate External Memory Manager

3.12.1. Boot usecase:

Core(s) Settings:

Context(s):	First Stage Boot Loader External Memory Loader
-------------	---

Boot:

Select boot code generation	true *
Selection of the boot system	Execute In Place
Header size	0x400 *

XIP:

select the memory	Memory 1
Application offset	0x100000 *

3.12.2. Memory 1:

Core(s) Settings:

Context(s):	First Stage Boot Loader
-------------	-------------------------

External Memory Loader

Select driver:

Select the memory driver

EXTMEM_NOR_SFDP

Configuration:

Memory Instance

XSPI2

Number of memory data line

EXTMEM_LINK_CONFIG_8LINES *

3.12.3. Memory 2:

Core(s) Settings:

Context(s):

First Stage Boot Loader
External Memory Loader

Select driver:

Select the memory driver

NONE

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context
DEBUG	PA14(JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	SWCLK	First Stage Boot Loader Application
	PA13(JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	SWDIO	First Stage Boot Loader Application
HDP	PC10	HDP_HDP2	Alternate Function Push Pull	No pull-up and no pull-down	High		First Stage Boot Loader
	PC4	HDP_HDP6	Alternate Function Push Pull	No pull-up and no pull-down	High		First Stage Boot Loader
	PC11	HDP_HDP3	Alternate Function Push Pull	No pull-up and no pull-down	High		First Stage Boot Loader
	PC9	HDP_HDP1	Alternate Function Push Pull	No pull-up and no pull-down	High		First Stage Boot Loader
	PC12	HDP_HDP4	Alternate Function Push Pull	No pull-up and no pull-down	High		First Stage Boot Loader
	PC7	HDP_HDP7	Alternate Function Push Pull	No pull-up and no pull-down	High		First Stage Boot Loader
	PD12	HDP_HDP5	Alternate Function Push Pull	No pull-up and no pull-down	High		First Stage Boot Loader
	PC8	HDP_HDP0	Alternate Function Push Pull	No pull-up and no pull-down	High		First Stage Boot Loader
LPUART1	PE5	LPUART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low		First Stage Boot Loader Application
	PE6	LPUART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	VCP_RX	First Stage Boot Loader Application
XSPI2	PN4	XSPIM_P2_IO2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	OCTOSPI_IO2	First Stage Boot Loader Application
	PN6	XSPIM_P2_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	OCTOSPI_CLK	First Stage Boot Loader Application
	PN8	XSPIM_P2_IO4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	OCTOSPI_IO4	First Stage Boot Loader Application
	PN0	XSPIM_P2_DQS0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	OCTOSPI_DQS	First Stage Boot Loader Application
	PN3	XSPIM_P2_IO	Alternate Function Push	No pull-up and no pull-down	Very High	OCTOSPI_IO1	First Stage Boot

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context
		1	Pull				Loader Application
	PN5	XSPIM_P2_IO3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	OCTOSPI_IO3	First Stage Boot Loader Application
	PN1	XSPIM_P2_NCS1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	OCTOSPI_NCS	First Stage Boot Loader Application
	PN9	XSPIM_P2_IO5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	OCTOSPI_IO5	First Stage Boot Loader Application
	PN2	XSPIM_P2_IO0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	OCTOSPI_IO0	First Stage Boot Loader Application
	PN10	XSPIM_P2_IO6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	OCTOSPI_IO6	First Stage Boot Loader Application
	PN11	XSPIM_P2_IO7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	OCTOSPI_IO7	First Stage Boot Loader Application
Single Mapped Signals	PC1	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	I2C1_SDA	
	PH9	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	I2CA_SCL	
	PB0	DEBUG_TRACED1	n/a	n/a	n/a	TRACED1	
	PB7	DEBUG_TRACED3	n/a	n/a	n/a	TRACED3	
	PB6	DEBUG_TRACED2	n/a	n/a	n/a	TRACED2	
	PB3	DEBUG_TRACECLK	n/a	n/a	n/a	TRACECLK	
	PE3	DEBUG_TRACED0	n/a	n/a	n/a	TRACED0	
	CSI_REXT	CSI_REXT	n/a	n/a	n/a		
	PB11	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	I2C2_SDA	
	PA15(JTDI)	DEBUG_JTDI	n/a	n/a	n/a	JTDI	
	PN12	XSPIM_P2_NCS2	Alternate Function Push Pull	No pull-up and no pull-down	Very High		
	PA10	ADC1_INP11	Analog mode	No pull-up and no pull-down	n/a		
	CSI_CKP	CSI_CKP	n/a	n/a	n/a		
	CSI_D1P	CSI_D1P	n/a	n/a	n/a		
	CSI_D0P	CSI_D0P	n/a	n/a	n/a		

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label	Context
	PA11	ADC1_INP12, ADC1_INN11	Analog mode	No pull-up and no pull-down	n/a	UCPD1_VSENSE	
	CSI_CKN	CSI_CKN	n/a	n/a	n/a		
	CSI_D1N	CSI_D1N	n/a	n/a	n/a		
	CSI_D0N	CSI_D0N	n/a	n/a	n/a		
	PB10	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	I2C2_SCL	
	PB5(JTDO/TRACESWO)	DEBUG_JTDO-SWO	n/a	n/a	n/a	SWO	
GPIO	PD10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	PWR_EN	First Stage Boot Loader* Application
	PD2	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	UCPD1_INT	First Stage Boot Loader* Application
	PC13	GPIO_Input	Input mode	No pull-up and no pull-down	n/a		First Stage Boot Loader* Application
	PO5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	CAM_NRST	First Stage Boot Loader* Application
	PO1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GREEN_LED	First Stage Boot Loader* Application
	PA5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	UCPD1_ISENSE	First Stage Boot Loader* Application
	PA7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	UCPD_PWR_EN	First Stage Boot Loader* Application
	PG0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED1_GREEN	First Stage Boot Loader* Application
	PG10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low		First Stage Boot Loader* Application
	PG8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low		First Stage Boot Loader* Application
	PN7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	USB1_OCP	First Stage Boot Loader* Application

* Initialized context

4.2. GPDMA1

4.2.1. Core(s) Settings:

Context(s):	First Stage Boot Loader Application
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4.3. HPDMA1

4.3.1. Core(s) Settings:

Context(s):	First Stage Boot Loader Application
-------------	--

4.4. LINKEDLIST

4.4.1. Core(s) Settings:

Context(s):	First Stage Boot Loader Application
-------------	--

4.5. NVIC configuration

4.5.1. NVIC1

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
Secure fault	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
FPU global interrupt	unused		
RCC secure global interrupt	unused		
LPUART1 global interrupt	unused		
XSPI2 global interrupt	unused		
IAC interrupt	unused		

4.5.2. NVIC1 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
Secure fault	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

4.5.3. NVIC2

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0

Interrupt Table	Enable	Preenmption Priority	SubPriority
Undefined instruction or illegal state	true	0	0
Secure fault	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
FPU global interrupt	unused		
RCC secure global interrupt	unused		
GPU2D cache interrupt	unused		
LPUART1 global interrupt	unused		
XSPI2 global interrupt	unused		

4.5.4. NVIC2 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
Secure fault	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Category viewContext Execution view

Choose filters ...

... by Context Execution

☐ First Stage Boot Loader

☐ Application

☐ External Memory Loader

Middleware

EXTMEM_MAHAGER

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Utilities	Bsp	Other
CORTEX_M55_...			LPUART1		RIF		DEBUG	PWR	LINKEDLIST		
CORTEX_M55_S			XSPI2				HDP				
GPDMA1			XSPIIM								
GPIO											
HPDMA1											
ICACHE											
HWIC1_S_Appli...											
HWIC_FSBL											
RCC											

5.2. Context Execution view

Category view Context Execution view

External Memory Loader

EXTMEM_MANAGER ✓

GPIO ⚠

RCC ✓

XSPI2 ✓

XSPIM ✓

First Stage Boot Loader

DEBUG ✓

GPIO ⚠

RCC ✓

XSPIM ✓

HDP ✓

IVIC_FSBL ✓

EXTMEM_MAH... ✓

LPUART1 ✓

XSPI2 ✓

CORTEX_M55_FS... ✓

ICACHE ✓

PWR ✓

Application

DEBUG ✓

GPIO ⚠

LPUART1 ✓

RCC ✓

XSPI2 ✓

XSPIM ✓

CORTEX_M55_S ✓

IVIC1_S_Application ✓

RIF ✓

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32n6-bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32n6-ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32n6-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32cubeai_press_pres.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32n6-series-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/st-neural-art-accelerator-introduction.pdf
Presentations	https://www.st.com/resource/en/product_presentation/presentation-st-edge-ai-suite.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32n6.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-

guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-

usb-typec-power-delivery-for-stm32-mcus-and-mpus-
stmicroelectronics.pdf

- Application Notes https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-with-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4861-introduction-to-lcdftt-display-controller-ltcd-on-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4908-getting-started-with-uart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5879-introduction-to-i3c-for-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5050-getting-started-with-octospi-hexadecapi-and-xspi-interface-on-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/cd00211314-how-to-

	optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5020-introduction-to-digital-camera-interface-dcml-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5967-getting-started-with-hardware-development-for-stm32n6-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5946-how-to-optimize-lowpower-modes-on-stm32n6-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an6000-how-to-build-the-discrete-power-supply-for-stm32n6-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an6197-stm32cube-mcu-package-examples-for-stm32n6-mcus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbp-d-sink-application-with-stm32cubemx-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5426-migrating-

for related Tools & Software	graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6197-stm32cube-mcu-package-examples-for-stm32n6-mcus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0620-stm32n6xxxx-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm01125716.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0273-stm32-cortexm55-mcus-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0486-stm32n647657xx-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf

Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf
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